General Toxicology Management

Primary Survey

General Impression
- **Assess environment for scene safety and exposure risks** (e.g., fumes, spills, paraphernalia, multiple patients)
- Look for signs of distress: agitation, respiratory depression, altered mental status, seizures

Medical: A - B - C
Airway, Breathing, and Circulation

Manage life threats as they are identified

Assess GCS & Neurological Status

Assess GCS: If <13, treat as high-risk
- Pupils:
  - Pinpoint: Consider: opioids, organophosphates
  - Dilated: Consider: stimulants, withdrawal
  - Unequal: Consider: Possible stroke or brain herniation
- Seizures or muscle twitching? → suggests TCA, isoniazid, or withdrawal
- Observe for agitation, hallucinations, or bizarre behavior (anticholinergic or hallucinogen)

Check Glucose

- For Altered Mental Status

Consider ALS Back-Up

Toxidrome Assessment

Toxidrome	Common Signs	Examples
Opioid	CNS depression, miosis, ↓RR, ↓HR	Heroin, morphine, fentanyl
Anticholinergic	Dry skin, hyperthermia, delirium, tachycardia, dilated pupils	Antihistamines, TCAs, atropine
Sympathomimetic	Agitation, tachycardia, hypertension, diaphoresis, dilated pupils	Cocaine, amphetamines, MDMA
Cholinergic	SLUDGE: salivation, lacrimation, urination, diarrhea, GI upset, emesis	Organophosphates, nerve agents
Sedative-hypnotic	CNS depression, normal pupils, hypotension, bradycardia	Benzodiazepines, barbiturates

Administer Specific Treatment

- Administer specific treatment as per relevant CPG (if applicable)
- If no specific CPG, manage as per Universal Care CPG.

Patient Care Goals

1. Rapid recognition of toxic exposure and early supportive management
2. Maintain airway, breathing, and circulation
3. Minimize toxin absorption when appropriate (e.g., decontamination)
4. Identify high-risk features requiring ALS backup or early transport
5. Provide early notification and accurate history to hospital team.
6. Ensure patient safety and prevent aspiration or secondary harm
7. Monitor and reassess regularly to detect deterioration earl

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Suspected or confirmed ingestion, inhalation, injection, or dermal exposure to a toxin, including;
  - Medications, and/
  - Naturally occurring or manufactured substances.
- Intentional or unintentional poisoning.
- Symptoms consistent with toxic exposure (e.g., altered mental status, seizures, bradycardia, respiratory depression).

Exclusion Criteria

- Isolated opioid overdose (see Opioid Overdose CPG)
- Isolated alcohol intoxication (see Suspected Alcohol intoxication CPG)
- Decontamination of patients after hazardous material exposure. (Requiring decontamination team managed by Civil defense).

Key Patient Assessment Points

1. **General Impression**
   - Observe level of consciousness, breathing pattern, and behavior on approach
   - Assess environment for scene safety and exposure risks (e.g., fumes, spills, paraphernalia, multiple patients)
   - Look for signs of distress: agitation, respiratory depression, altered mental status, seizures
2. **Airway, Breathing & Circulation (ABC)**
   - Airway:
     - Is the airway patent? Is the patient protecting it?
     - Suction needed? Risk of aspiration?
   - Breathing:
     - Count RR, assess for shallow or irregular breathing
     - Snoring or slow respirations may indicate opioid or sedative overdose
     - Tachypnea may indicate metabolic acidosis (e.g., salicylates, methanol)
   - Circulation:
     - HR and BP: Bradycardia → possible beta-blocker or CCB; Tachycardia → possible anticholinergic, sympathomimetic
     - Capillary refill, skin color, and temperature
     - Check for arrhythmias, wide QRS, or prolonged QT
3. **Disability / Neurological Status**
   - Assess GCS: If <13, treat as high-risk
   - Pupils:
     - Pinpoint: Consider: opioids, organophosphates
     - Dilated: Consider: stimulants, withdrawal
     - Unequal: Consider: Possible stroke or brain herniation
   - Seizures or muscle twitching? → suggests TCA, isoniazid, or withdrawal
   - Observe for agitation, hallucinations, or bizarre behavior (anticholinergic or hallucinogen)
4. **Exposure & Skin Assessment**
   - Inspect for:

    - Track marks or injection sites
    - Rashes or blistering (e.g., chemical exposure)
    - Sweating (cholinergic, sympathomimetic) vs dry skin (anticholinergic)
  - Remove contaminated clothing if dermal toxin suspected
  - Check temperature: hyperthermia suggests sympathomimetic or serotonin syndrome
5. **Focused Toxidrome Assessment**
  - Look for pattern-based clues ("toxidromes"):

Toxidrome	Common Signs	Examples
Opioid	CNS depression, miosis, ↓RR, ↓HR	Heroin, morphine, fentanyl
Anticholinergic	Dry skin, hyperthermia, delirium, tachycardia, dilated pupils	Antihistamines, TCAs, atropine
Sympathomimetic	Agitation, tachycardia, hypertension, diaphoresis, dilated pupils	Cocaine, amphetamines, MDMA
Cholinergic	SLUDGE: salivation, lacrimation, urination, diarrhea, GI upset, emesis	Organophosphates, nerve agents
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6. **Blood Glucose Check**
  - Mandatory for all patients with altered mental status
  - Hypoglycemia may mimic or complicate overdose
  - If abnormal, refer to Hypoglycemia or Hyperglycemia CPG
7. **SAMPLE History (Targeted for Toxicology)**
  - S (Symptoms): When did they start? Any seizures, vomiting, hallucinations?
  - A (Allergies): Drug allergies, previous reactions
  - M (Medications): Current prescriptions, over-the-counter, supplements, recreational substances
  - P (Past history): Psychiatric illness, substance use, diabetes, renal/liver failure
  - L (Last oral intake): What, how much, and when? Was alcohol involved?
  - E (Events): Suicide attempt, peer pressure, unknown ingestion, multiple patients affected?
8. **Determine Triage Category**
  - Determine Triage Category as per Patient Triage Categories CPG.
    - Red Priority: GCS <9, RR <10, seizures, bradycardia/hypotension, pediatric ingestion
    - Yellow Priority: Mild symptoms, GCS 9–13, isolated vomiting
    - Green Priority: Alert, asymptomatic, known non-toxic ingestion

Patient Management & Interventions

1. **Consider ALS back-up**
  - Consider the requirement for ALS back-up as per Universal Care CPG.
  - ALS back-up should be routinely requested for:
    - GCS <13 or rapidly dropping
    - Seizures, dysrhythmias, or cardiovascular instability
1. **Decontamination**
  - Remove contaminated clothing and irrigate skin/eyes with water if dermal exposure suspected.
  - Note: Patient decontamination of harmful substances should be performed by Civil Defense.
2. **Primary Survey**
  - Perform Primary Survey as per Universal Care CPG.
3. **Oxygen Therapy**
  - High-flow oxygen for altered LOC, respiratory distress, or suspected CO poisoning
  - Target SpO_2 ≥94%
4. **Specific Therapy / Medication Administration**
  - If Opioid overdose suspected go to Opioid Overdose CPG.
  - If Calcium Channel Blocker overdose suspected go to Calcium Channel Blocker Overdose CPG.

- If Beta Blocker overdose suspected, go to Betablocker Overdose CPG.
- If Organophosphate poisoning suspected, go to Organophosphate Poisoning CPG.

5. **IV Access**
   - Obtain IV access if patient is hypotensive, vomiting, or unconscious
6. **Fluid Administration**
   - If signs of shock present, administer 10–20 mL/kg 0.9% NaCl bolus
7. **ECG Monitoring**
   - Monitor for bradycardia, wide QRS, arrhythmias
8. **Patient Reassessment**
   - Frequently Reassess the patient as per Universal Care CPG.
   - Specific Patient Reassessment Points:
     - Pupil size and reactivity
     - Presence of seizures, twitching, or new neurological symptoms
     - Patient response to interventions (e.g., oxygen, fluids, naloxone)
     - Signs of deterioration (e.g., decreasing GCS, new vomiting, irregular breathing)
     - Scene and crew safety, particularly after reversal agents are used
9. **Transport & Early Notification**
   - Transport to hospital is required for all symptomatic patients
   - Notify hospital if ingestion of cardiotoxic agents or pediatric patient with possible toxic ingestion

Patient Safety Considerations

- Wear gloves and PPE with unknown substance exposure
- Avoid triggering vomiting or giving anything orally unless directed
- Be prepared for airway compromise even in initially stable patients
- Pediatric patients can get worse very quickly; when unsure, treat the situation seriously and act early
- Avoid judgmental questioning; treat as a medical emergency

Clinical Insights & Additional Guidance

- Many overdoses are mixed; don't focus on one substance unless confirmed
- ECG changes may indicate sodium channel or potassium channel toxicity (e.g., TCA, lithium)
- In pediatric ingestions, even small amounts can be lethal (e.g., calcium channel blockers, opioids, hydrocarbons)
- Consider scene safety: hazardous substances, paraphernalia, agitated patients
- Consult TeleEMS early in ambiguous or critical cases

Documentation

- Identification: Age, weight (if pediatric), sex
- Medical complaint: Toxic exposure, overdose, suspected ingestion
- Injuries/Information: Type of toxin, route, amount, timing
- Signs: GCS, HR, RR, BP, pupils, SpO_2
- Treatment: Oxygen, IV access, fluids, airway management
- Allergies: Drug allergies
- Medications: Chronic meds or recent psych/analgesic drugs
- Background: Psychiatric history, previous overdose, renal/liver disease
- Other: Suicidal intent, family present, police notified

Key Performance Indicators

1. Investigate and document substance type, route, amount, and exposure/ingestion time.
2. Request ALS back-up for unstable or GCS <13 cases

References

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2. DCAS. (2024). *Request for Back-Up Policy (DCAS-AOS-QM-PL02-AR)*. Dubai Corporation for Ambulance Services.
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4. Greaves, I. and Porter, K. (2007). *Oxford Handbook of Pre-Hospital Care*. Oxford: Oxford University Press.
5. Nelson, L.S., Howland, M.A., Lewin, N.A., Hoffman, R.S., Goldfrank, L.R. (2019). *Goldfrank's Toxicologic Emergencies*. 11th ed. New York: McGraw-Hill Education.
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